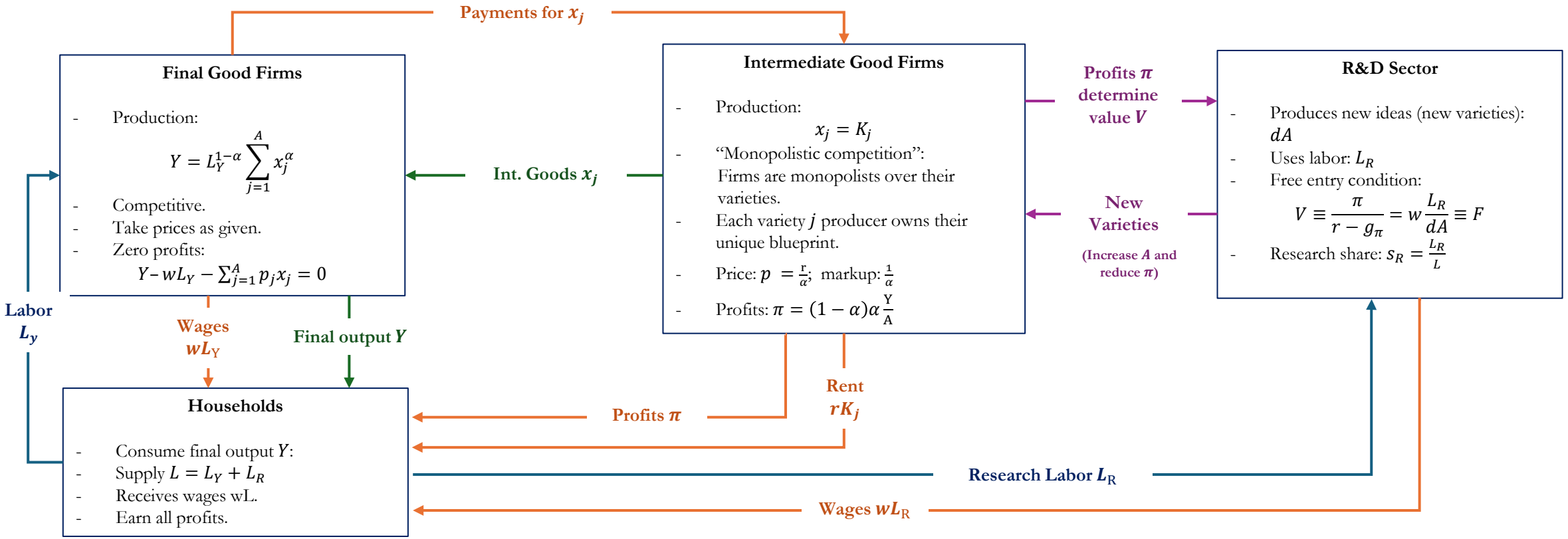
Romer Endogenous Growth Model with R&D and Microfoundations*



Mechanism

- ▶ Market Power
  - ▶ Profits
    - ▶ Entry
      - ▶ Varieties
        - ▶ Growth

Aggregate Outcomes

- Symmetric eq: $p_j = \frac{r}{\alpha}; x_j = \frac{K}{A}$
- Aggregate supply: $Y = L_Y^{1-\alpha} A x^{\alpha} = K^{\alpha}(AL_Y)^{1-\alpha}$
- GDP from income point of view:

$$Y = \underbrace{wL_y}_{(1-\alpha)Y} + \underbrace{\underbrace{A\pi}_{(1-\alpha)\alpha Y} + \underbrace{rK}_{\alpha^2 Y}}_{\alpha Y}$$

Dynamic Macro Relationships

- Profit growth grows with market size: $g_\pi = g_L$
- Technology growth: g_A
- Balanced Growth Path: $g_Y = g_L + g_A$

*Summary diagram prepared for Econ 414 – Winter 2026

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